Depth, Water, and Oak Gall

Living ink, next law. Seven changes: paper depth and dents, ink in depth with a side view, the water brush, the oak gall color, fading by exposure, paper that remembers water, and splatter. Build spec for the coding agent.

The principle all four follow

Nothing computes unless it is touched or viewed. A tile runs only while it has wet ink. Time is applied to a record only when someone looks at it, in one step, never tick by tick while nobody is there.

This is the same rule as the sleep fix for the long-stroke lag: dry tiles leave the active set and only a new deposit wakes them. Aging follows it too, with two clocks matching their real causes: **air by time** (the oak gall color darkens with elapsed time) and **light by being seen** (every color fades with exposure, counted in views). Both are applied in one step at view time. A sheet nobody opens costs nothing and does not change.

Law version

Law v1 is frozen. All four changes go into `law/v2/`, a new frozen package with its own goldens. Records carry their law number: v1 records keep rendering under v1 exactly as before, and new writing is made under v2. Nothing about a v1 sheet changes when v2 ships.

1. Paper depth and dents

The paper gets a short Z axis. It stays a 2D grid of cells, but each cell now has a surface height that writing can push down.

- **Dent.** Each cell gets a new field `dent` (Uint8, starts at 0). On deposit, the nib footprint pushes the surface down by an amount from pressure × dwell: `dent += DENT_Q[p] × dt`, clamped to `DENT_MAX`. Plastic: a dent never recovers. With no real pressure, synthetic flex p and dwell drive it, as for width.
- **A dent is a groove.** It feeds into flow two ways: it lowers the cell's height in ϕ (so surface ink runs into it), and it raises the cell's capacity (`cap += dent × DENT_CAP`, so the groove holds more ink). Real paper does this: scratched lines pool ink.
- **Fibers compress.** In a dented cell, sizing (how fast it drinks) drops a little, so ink in a groove sits wet longer and settles darker.
- **It is a fact.** The dent map is part of the settled state and goes into `factsDigest`. It comes only from the acts and the paper, so it regrows byte for byte like everything else.

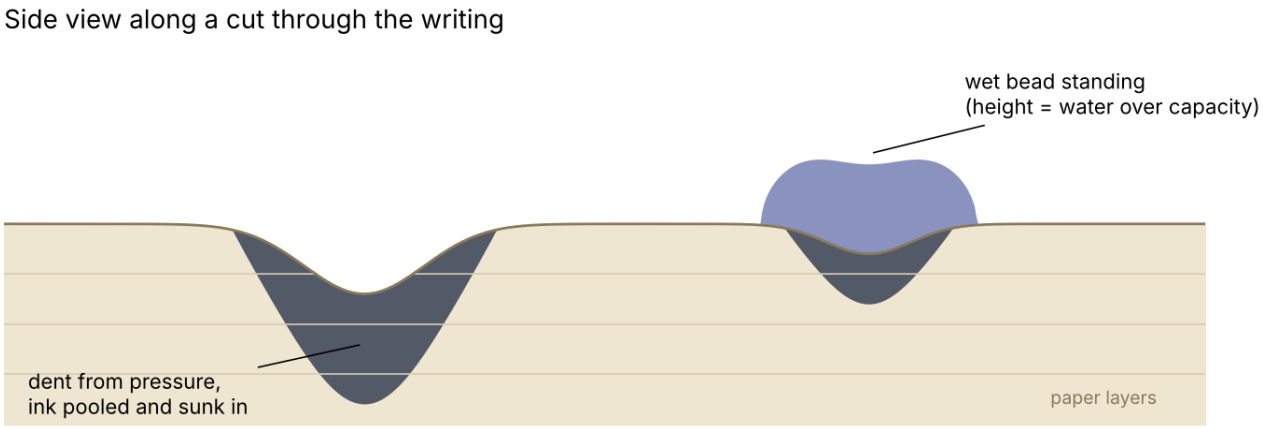
2. Ink in depth: beads, sinking, and the side view

The sim already has height: wet water over capacity is the bead. Two things are added so depth can be seen.

- **Penetration.** Each inked cell gets `depth` (Uint8): how far settled ink sank into the fibers. During settle, `depth += SETTLE_DEPTH[siz] × settled_amount`, clamped to the paper thickness. Thirsty cells and grooves sink deeper.
- **Peak height.** Each inked cell records `peak` (Uint8): the highest the wet bead stood over it before drying. That keeps the droplet visible in the side view after it has dried.
- Both are facts, in the digest.

The side view

A viewer mode, not part of the published look. Pick a line across the writing; the view shows that cut: the paper body in layers, the surface with its dents, ink sunk into the fibers below the surface, and the bead above it (live while wet; from `peak` after drying).



The relief view

The top view lit from one side, using dent and bead height as a height map, so grooves and dried droplets read as depth on the page. Also a viewer mode. The published look stays flat, as ruled before (no height shading in published looks).

- Both views read the same facts, so they work for any sheet the viewer can see: own and open sheets regrow on the device, and closed sheets are served by the render pool like looks (`/look?mode=side&cut=...` and `mode=relief`).

3. The water brush

A pen kind that carries water and no pigment. Use it over dried ink and the ink comes back to life and bleeds again.

- **Pen.** `kind = water` in the pen act. Deposit adds W only; pigment supply is 0. Same nib shapes, same flow and dwell behavior.
- **The react: water + fixed ink → wet ink.** In any cell with wet water and fixed pigment, each dry tick returns a fraction of the fixed ink to wet: ΔP `= F × REDISSOLVE[ink] × W / W_SAT`. Then normal flow carries it, so the old line feathers and blooms where the water spread.
- **Oak gall resists.** Ordinary color revives at `REDISSOLVE`. Gall revives at the same rate while pale, and less as it darkens (`REDISSOLVE × (1 − dark)`), true to life.
- **Waking.** A water deposit wakes the tile like any deposit. Dormant tiles that hold fixed ink come back into the active set, run until dry, and sleep again.
- **Scope in this build:** the water brush works on writing in the current draft. Reviving ink on an already sealed sheet needs a new record that starts from the old sheet's facts (see Needs ruling).

4. The oak gall color

There is one ink. Every color in the palette behaves the same way it does now, with one exception: **oak gall**, a single chosen shade of brownish, greyish, bluish black. That hex is the code for ink that ages. It goes down paler and darkens toward its shade as it meets the air.

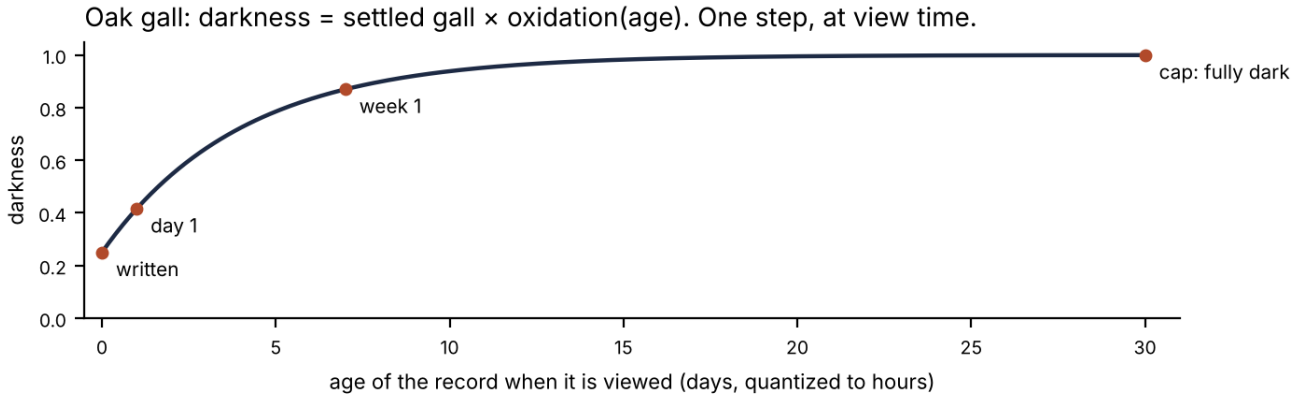
- **Palette entry:** `OAK_GALL = #353438` (default, Tabor's to set). Shown in the color options like any other color.
- **Its own channel.** Oak gall does not write into the RGB pigment channels. It writes into one gall channel `G` (wet) and `FG` (fixed), allocated only on tiles that have ever held gall. It flows, mixes position-wise, settles, and dents exactly like any ink.
- **Shown absorbance** per cell = RGB pigment + `FG × ABS(OAK_GALL) × dark`. Mixed with another color, the other color stays as it is and the gall darkens only where gall reached.
- **Fresh tint.** At darkness 0 it shows at `GALL_FRESH` of its strength (a pale grey-blue), so it is visible while writing.

What darkens it: air, by time

Oak gall darkens because oxygen from the air reacts with it. Air is everywhere and unlimited, so it is a constant of the law, not matter with cells. What varies is each cell's access to the air, and the cell's own facts decide that:

```
access[cell] = AIR[depth, thickness]     sunk deep or under a thick pool = less air
rate[cell]   = OX_BASE × access[cell]
dark(age)    = OX[ age × rate[cell] ]     literal table 1 − e^(−x), capped at AGE_CAP
age          = view_time − record.made, quantized to hours
```

- Hairlines darken first; pools and grooves lag, then end up deepest. Every gall letter darkens unevenly, for real reasons, and the same letter always darkens the same way.
- **Zero cost while unviewed.** Nothing is stored or scheduled. One pass at view; a year costs the same as a day.
- **Still exact.** `FG`, depth and thickness are sealed facts with no time in them; verification never depends on when it is checked. The look for a given age bucket is a pure function of the facts and that bucket.



5. Fading: light, by being seen

Real ink ages more from light than from years: a letter kept in a drawer for two centuries can look fresh; one displayed for twenty looks ancient. Here, light is viewing. Every color, oak gall included, fades with how much the record has been seen.

- **Exposure** is a counter per record: the number of views, counted at most once per viewer per day, so nobody can fade a page by refreshing. The owner's own views count too.
- **Fading is per cell and reads depth.** Light hits the surface first: `fade[cell] = FADE[ exposure × SURFACE[depth] ]`. Ink that sank deep or pooled in a dent survives longest, so a much-viewed page keeps the strokes that pressed hardest, as real faded manuscripts do.
- **Shown** = shown absorbance × `fade[cell]`. Applied in the same view-time pass as darkening.

- **The counter is new data**, not a copy of anything, so it does not break the one-storage rule: `data/users/<id>/exposure.json` (and per letter), `{inkId: count}`, written through the per-user queue.
- **The look** becomes a pure function of three things: sealed facts, age bucket, exposure bucket. Cached per (ink, scale, age bucket, exposure bucket). Verification is unchanged: fading is how the record looks after being lived with, not part of what was written.

6. The paper remembers water

Paper changes when it gets wet, and the change stays after it dries. Soaked paper drinks faster and feathers more the next time; wet paper is weak and tears if you keep working it; heavily soaked paper dries buckled. All of this becomes paper facts, made only while a cell is wet, permanent after.

- **Soak.** A new per-cell field `soak` (Uint8) adds up how much water the cell has held: each dry tick, `soak += SOAK_Q × W / W_SAT`, clamped. It never goes down.
- **Sizing washes out.** The cell's effective sizing drops with soak (`siz_eff = siz − soak × SOAK_SIZ`) and its fibers loosen (fiber conductance rises with soak). Ink written later on a once-soaked area drinks in faster and feathers more. A heavy wash, dried, leaves a thirsty patch that shows on every line written over it.
- **Wet paper is weak.** While a cell is wet its strength drops: `strength = BASE − W × WET_K − soak × SOAK_K`. Each deposit pass over a wet cell adds `wear` (Uint8) by pressure (or synthetic pressure and dwell). Going over the same wet spot again and again pills it, and when `wear` passes `strength` the cell tears.
- **A tear is permanent.** A torn cell (`torn` bit) holds no ink, shows the page behind it in the look, appears as a break in the side view, and ink arriving at its edge runs to the torn rim and pools there.
- **Cockling.** As a heavily soaked region dries, the surface buckles: `z += COCKLE[soak]` as a low, smooth ridge pattern from the paper's own fibers. Later ink runs off the ridges into the valleys.
- **All of it is facts** (soak, wear, torn, the cockle in z), in the digest, regrown from the acts and the paper like everything else. It costs nothing on dry tiles: it is computed only inside the wet loop that already runs.
- **The water brush** is the natural way to do this on purpose: wash an area, let it dry, write over it and watch the ink spread differently.

7. Splatter

Real pens throw ink: a fine spray when the nib snags a fiber, drops flung on a fast turn, a few drops on a hard landing, a fat drop from an overloaded nib. There is no randomness in the law, so the chance comes from the paper.

- **Odds come from the moment.** At each act, an integer probability `P_SPLAT = f(speed, turn sharpness, carry/flow, pressure, paper tooth at the nib) × spatter`. Snag rises with speed on upstrokes over rough cells; flick with sharp turns at speed; impact with fast pen-down and high load; overload with dwell at high flow.
- **The paper decides.** Whether it fires: `H(paperKey, 'splat', act index) < P_SPLAT`. If it fires, the same hash stream gives the drop count, sizes (long tail: many specks, rarely a big drop) and offsets (along the motion, distance rising with speed; around the point for impact).
- **A drop is a deposit.** Each drop adds water and the current pigment at its point. Nothing else is new: the existing rules make it bead, ring, sink, feather, dent nothing (no pressure), and dry.
- **Facts, not decoration.** Splatters are in the settled facts and the digest. Same strokes on the same paper give identical splatters on every regrowth; the same strokes on another person's paper give different ones, so a tracer cannot copy them. Undo removes a stroke's splatters with it. Nothing is stored; they regrow from

the acts and the paper.

- **Pen setting:** `spatter` 0..N on the pen (0 = never). Flex nib defaults higher than fine nibs.

State and data cost

```
per cell, v1:      W, P0..2 (wet)   F0..2 (fixed)                        7 integers
per cell, v2 adds: dent, depth, peak, soak, wear (Uint8 each) + torn bit  +6 bytes
gall, sparse:      wet G and fixed FG, only on tiles that ever held gall   +2 on those tiles
per record:        exposure count (one integer, in exposure.json)
```

- Acts do not grow: the water brush is a pen kind and oak gall is a palette color, both already in PNKA's pen fields. A minute of writing is still about 20 KB.
- Nothing about depth or darkening is stored; it all regrows from the acts and the paper. The age is computed from the record's own `made` time.

Tests to add

- **Dent:** a hard slow stroke leaves a deeper dent than a light fast one; ink later laid across the groove pools in it (more fixed ink in the dented cells than beside them).
- **Depth:** thirsty cells sink deeper than sized cells; a dwell shows peak height in the side view after drying.
- **Water:** water over dried ordinary ink re-wets and bleeds (fixed decreases, wet appears, ink spreads past the old edge); water over darkened gall barely moves it; dormant tiles wake and go dormant again.
- **Oak gall mixing:** gall crossed with a blue line darkens only where gall reached; the blue is unchanged.
- **Uneven darkening:** at 1 day, a gall hairline is darker than a gall pool on the same sheet; at the cap, the pool is darker.
- **Lazy darkening:** the look at age 0, 1 day, 7 days, 30 days matches the table; 400 days equals the cap; facts digest identical at every age; no timer, job, or write happens for an unviewed gall sheet.
- **Fading:** exposure 0 equals the fresh look; high exposure fades surface ink more than dented or sunk ink; ten refreshes by one viewer in a day add one exposure; the facts digest is identical at every exposure.
- **Soak:** ink written over a washed-and-dried area spreads wider than the same stroke on fresh paper; soak never decreases.
- **Tear:** scrubbing one wet spot repeatedly tears it; the same strokes on dry paper do not; a torn cell holds no ink and ink pools at its rim.
- **Cockle:** a heavily soaked region dries with raised ridges, and later ink collects in the valleys.
- **Splatter:** spatter 0 never splatters; fast flicks splatter more than slow curves; identical strokes on the same paper give identical splatters; on different paper, different splatters; undo removes them; total ink accounting includes drop volume.
- **Sleep rule:** a 60-second stroke with dents, depth and soak still has flat per-tick cost.
- **Versions:** every v1 golden still renders identically under the v1 package.

Needs Tabor's ruling

- Water on sealed sheets: a new record that starts from an old sheet's facts (a continuation sheet), or water only within drafts.
- The oak gall hex (default #353438), darkening time τ, and the cap.
- Fading strength (how many views to visibly faded), and whether the exposure count is shown under a page.
- Whether the side view and relief view are shown to visitors, or only to the writer and on open sheets.

- Paper thickness (how many depth levels), and dent strength for mouse and thumb with no real pressure.
- How easily wet paper tears (how many scrubbing passes), and whether tearing is on by default or a paper-family setting (e.g. a heavy watercolor paper that barely tears).
- Default spatter level, and whether it is on by default.